5. The null hypothesis in the ANOVA is that all the population means are

(a) Equal (b) Variable

(c) Unequal (d) none of the above

6. In one-way classification with 30 observation and 5 treatments the degrees of freedom for error is

(a) 29 (b) 4 (c) 25 (d) 150

7. In two-way classification the total variation *TSS* is

(a) *SST*+*SSB*+*SSE* (b) *SST*-*SSB*+*SSE*

(c) *SST*+*SSB*-*SSE* (d) *SST*+*SSB*

8. In two-way classification if *TSS* = 210, *SST* = 32, *SSB* = 42 then *SSE* =

(a) 126 (b) 74 (c) 136 (d) 178

9. In two-way classification with 5 treatments and 4 blocks the degrees of freedom due to error is

(a) 12 (b) 19 (c) 16 (d) 15

10. The formula for comparing three or more means in one-way analysis of variance is

(a) $F = \frac{MST}{MSE}$ (b) $F = \frac{TSS}{SST}$

(c) $F = \frac{MSB}{MST}$ (d) $F = \frac{MST}{MSB}$

11. _____ test is used to compare three or more means.

(a) t (b) χ^2 (c) F (d) Z

12. When there is no significant difference among three or more means the value of F will be close to

(a) 0 (b) -1 (c) 1 (d) ∞

13. *F*-test is also called as

(a) mean ratio test (b) variance ratio test

(c) variance test (d) standard deviation ratio test

14. The Analysis of Variance procedure is appropriate for testing the equivalence of three or more population

(a) variances (b) proportions

(c) means (d) observations

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